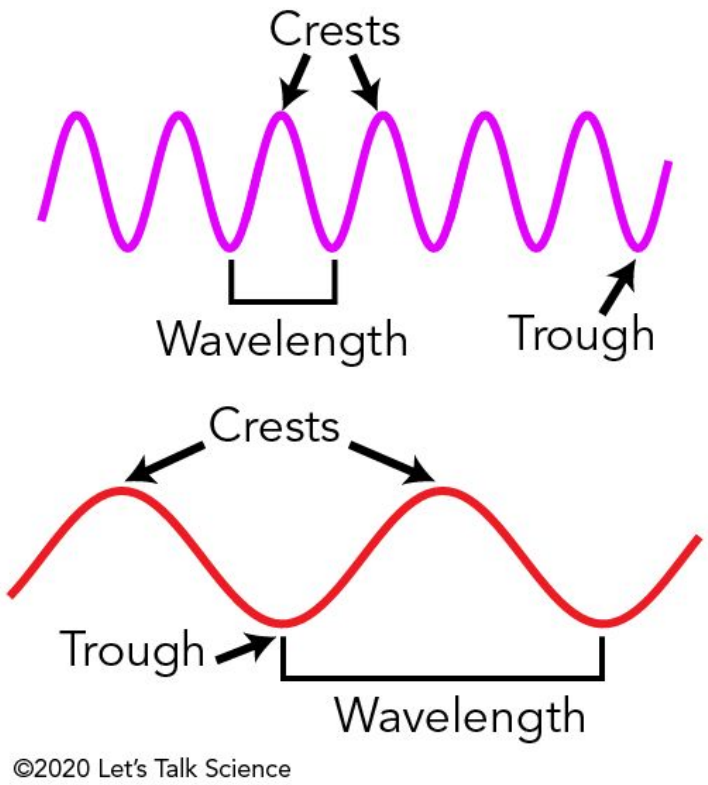
Light & Spectroscopy

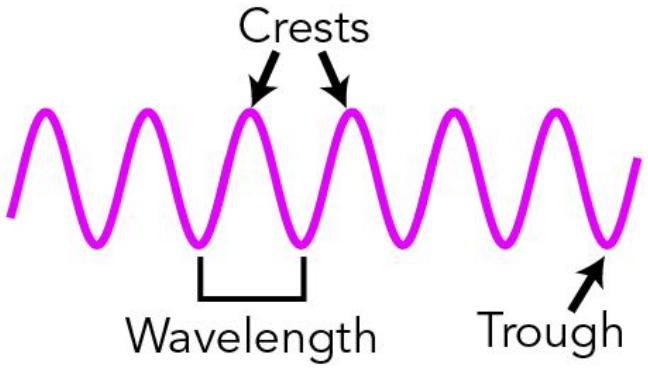
Waves

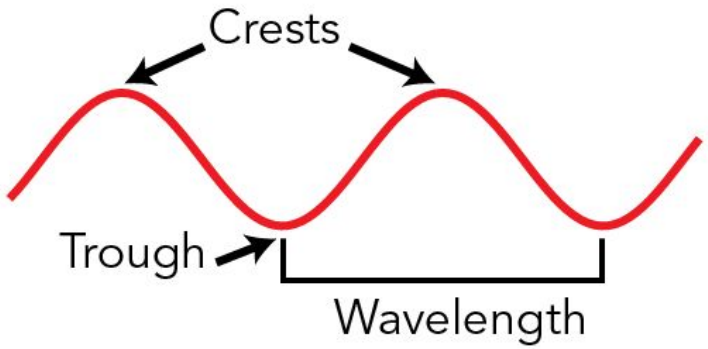
- Light has the properties of **waves**
- Waves transmit **energy** and travel through matter and empty space
- Different types of light are distinguished by their **wavelength**



Wavelength

- Like ocean waves, light waves have **crests** and **troughs**
- The **wavelength** of light is the distance measured from crest to crest (or trough to trough) in one of its waves





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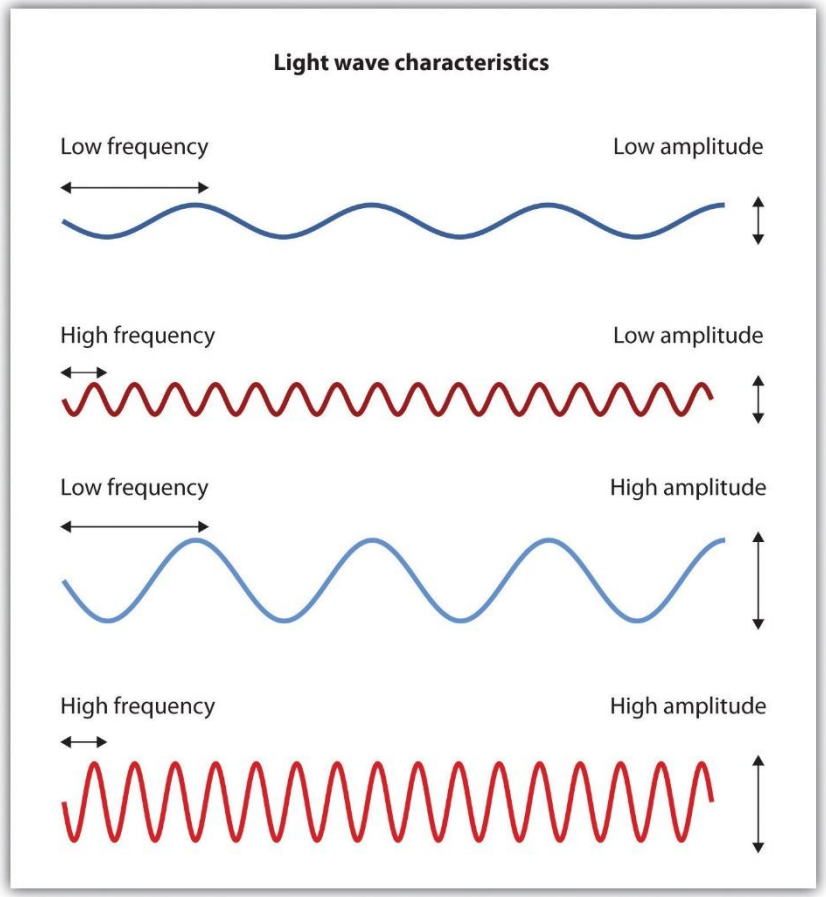
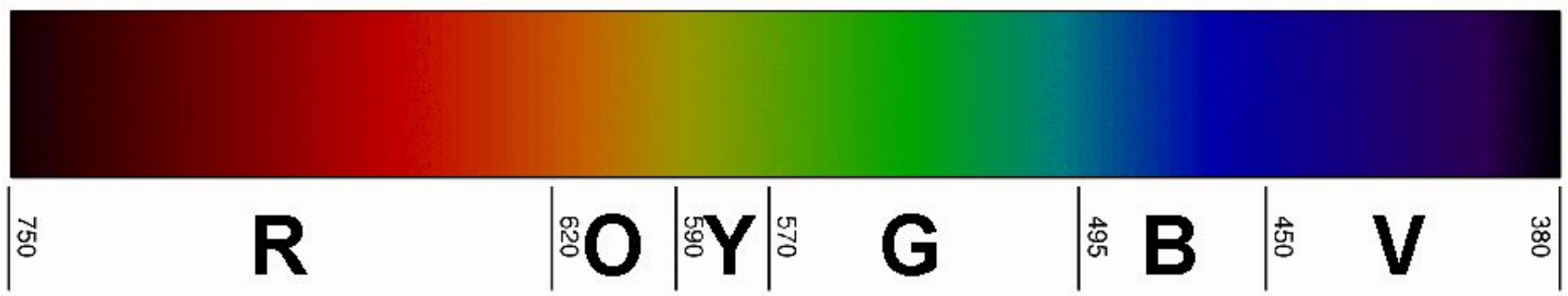
Frequency

- Another property of light related to wavelength is its **frequency**
- Frequency is the **number** of crests that pass by a given point every **second** as the wave moves



The Visible Spectrum

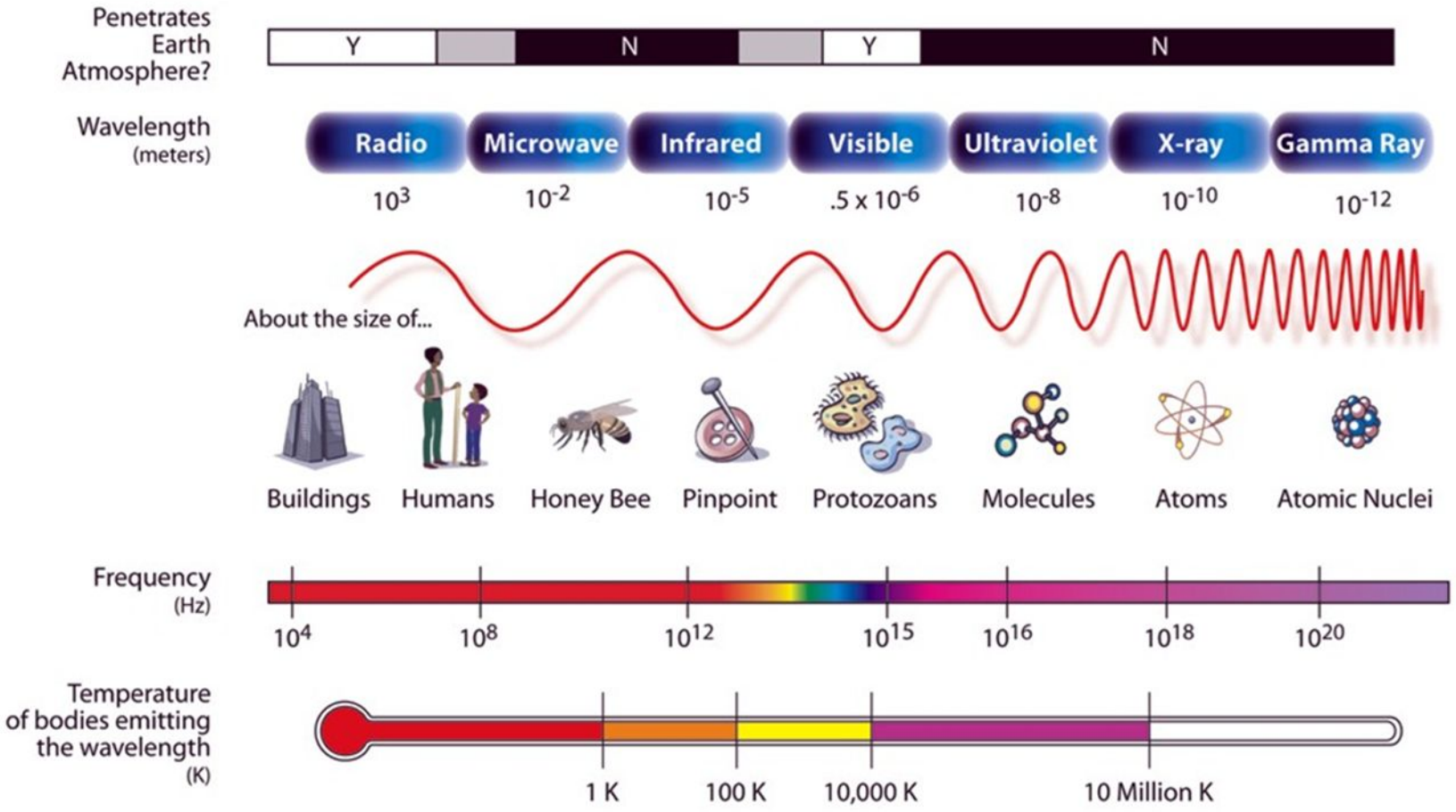
- The special band of light waves that the **human eye** can detect



The Electromagnetic Spectrum

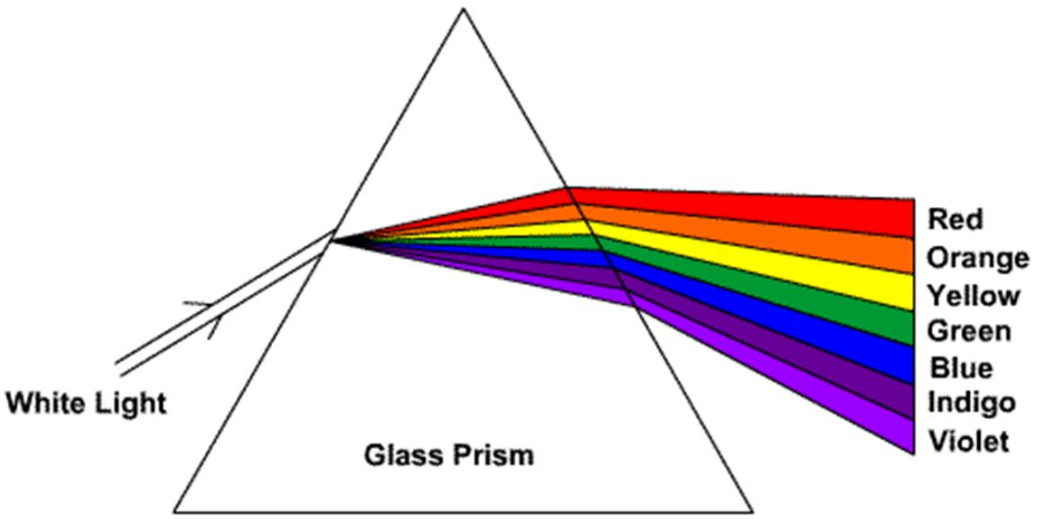
- This spectrum classifies electromagnetic waves (waves that carry both **electrical** and **magnetic** energy) based on the amount of **energy** they contain
- Their energy is transferred by **radiation** – meaning it doesn't need a medium to pass through

THE ELECTROMAGNETIC SPECTRUM



Spectroscopy

- Spectroscopy is the study of the **absorption** and **emission** of light and other radiation by matter
- It involves the splitting of light into its constituent wavelengths (a **spectrum**)

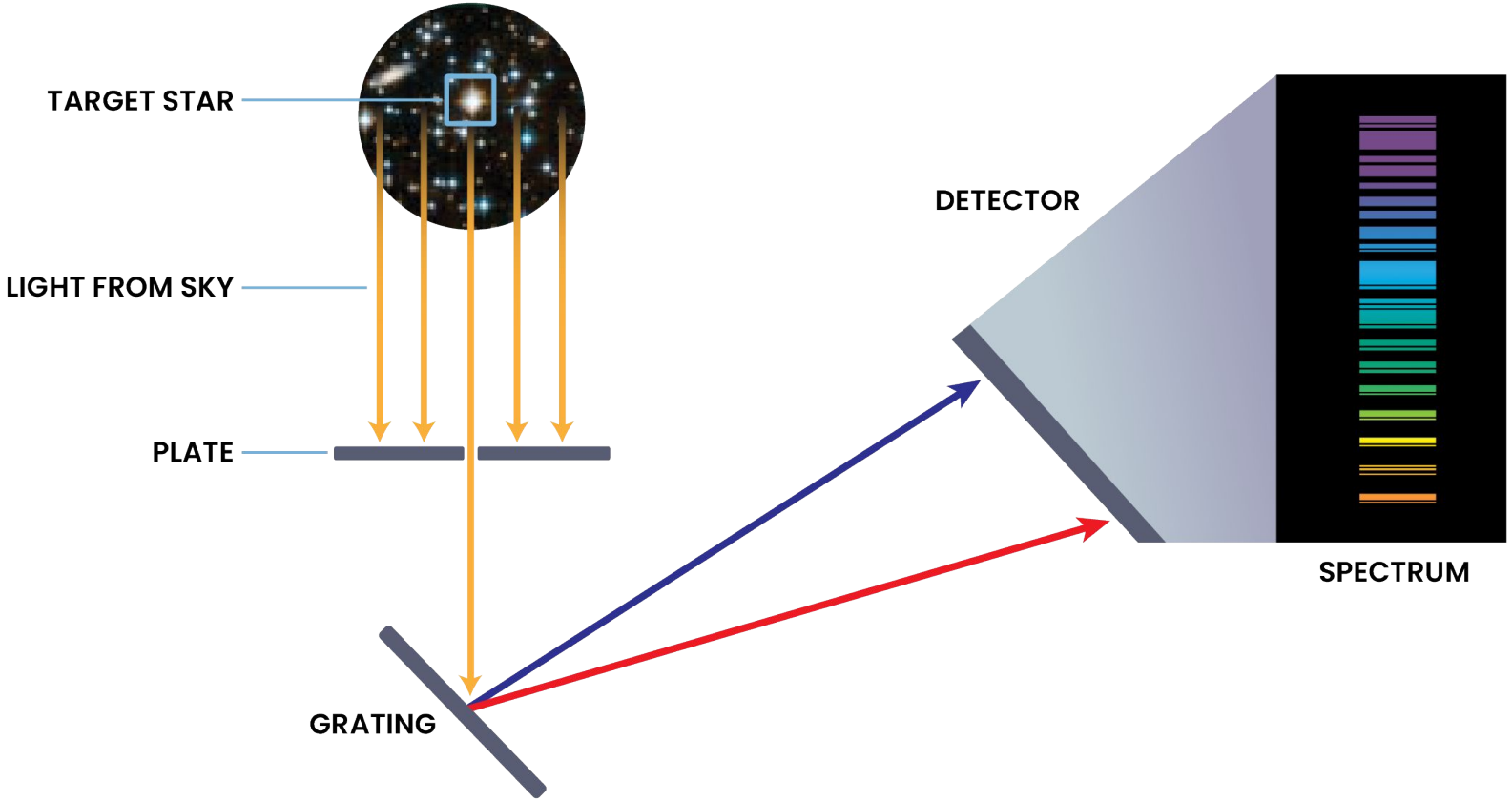


Spectroscope

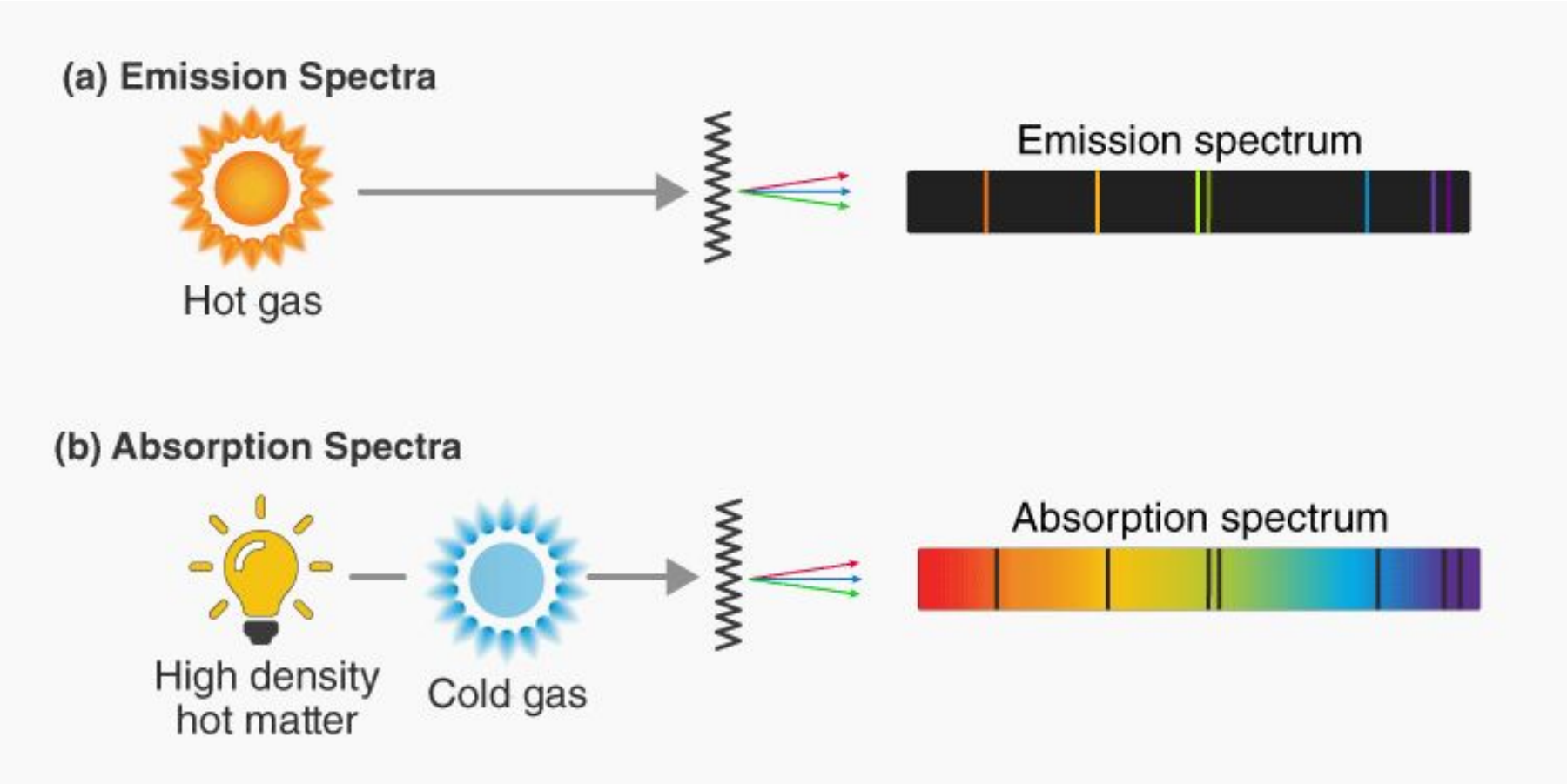
- A spectroscope or spectrometer passes light coming into the telescope through a tiny hole or **slit** in a metal plate to isolate light from a single area or object
- This light is bounced off a special **grating** , which **splits** the light into its different wavelengths (just like a prism makes rainbows)
- The split light lands on a **detector** , which records the spectrum that is formed

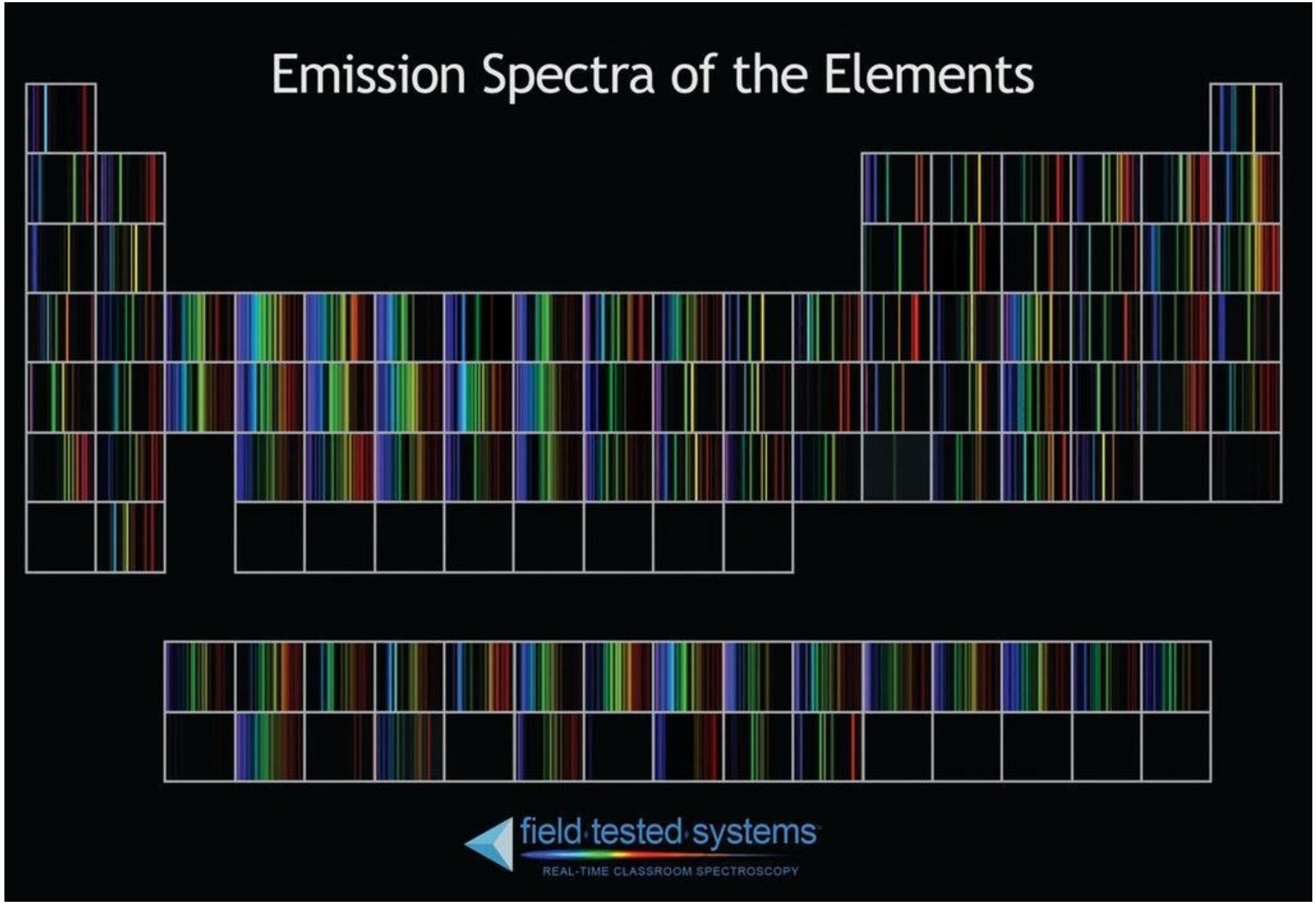
Spectroscopy explained

Spectroscope



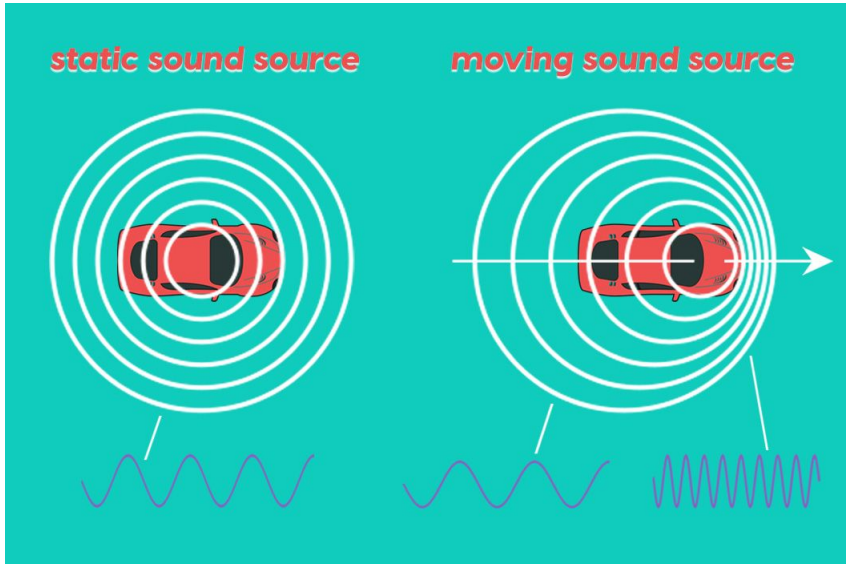
Line Spectra




Emission Spectra of the Elements
field·tested·systems
REAL-TIME CLASSROOM SPECTROSCOPY

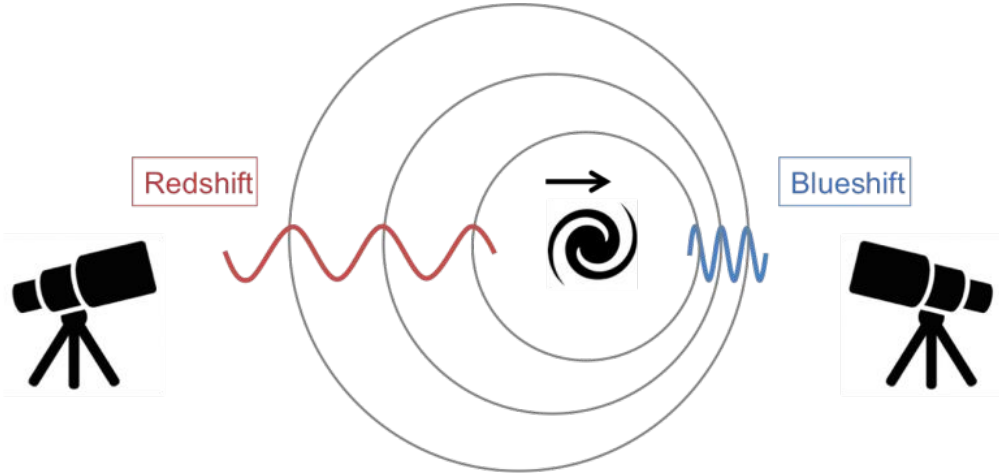
The Doppler Effect

- Imagine a car driving towards you blowing its **horn**
- As it drives towards you, the wavelengths are **compressed** and the resulting sound is a **higher** pitch (shorter wavelength)
- When it passes you and is moving away, the wavelengths are **stretched** and the pitch is **lower**



Shift

- If an object is moving **away** from us, the light waves are **stretched** out and the whole spectra experiences a shift towards the **red** side of the spectrum
- If an object is moving **towards** us, the light waves are **compressed** shift towards the **blue**



Your turn

1) Red Shift, Blue Shift
2) Analyzing Spectral Patterns/Mystery Spectra

<u>Catch up from previous lessons (if not already done):</u>

1) Northern Lights Questions
2) Astronomy 101